

Predicting the Sheltered Dogs' Likelihood of Adoption Using Logistic Regression Model

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Abstract

The study examines the factors affecting the adoption likelihood of sheltered dogs. By establishing a multiple linear regression from the dataset “pet_adoption_data.csv” that is obtained from Kaggle, factors such as breed, age, color, size, health status, and time in shelter days are used to predict the likelihood of adoption in a logistic model. When the numbers are calculated in the range of 0 to 1, they are then converted to binary numbers 0 and 1 to test for the accuracy of the model. The p-value of the model suggests that the breed and the size of the dog and the vaccination status has a significant influence on determining the likelihood of the adoption. While more in-depth investigation is needed, the finding can give a hint on how to utilize the resources in the dog shelters to make the maximize the adoption rate.

1. Introduction

In America, the first animal shelter was built on April 14, 1869, by a group of women activists in Bensalem (Reynolds, 2021). The group's idea was to “provide as soon as possible, a Refuge for lost and homeless dogs, where they could be kept until homes could be found for them, or they be otherwise disposed of” (Reynolds, 2021). The idea of animal support was further expanded in the late nineteenth century. Around that time, people started to root for social reforms “for the human treatment of factory workers” and the humane treatment of the animals naturally followed the flow (Reynolds, 2021).

Adopting the dog from a shelter has been considered an ethical choice as it prevents dogs from being euthanized. The ultimate aim for the shelters is to minimize the length of stay (LOS) by branding the dogs appealing to the customers as the lesser the LOS is, the more chance to “maximize adoptions, animal welfare, and reduce crowding” (Mohan-Gibbons, Weiss, Garrison, & Allison, 2014). Despite the effort of the workers, in a paper written by Mohan-Gibbons et al. (2014), there’s an estimation that each year “millions of dogs enter the sheltering system in the US” and “of those ... as many as 60% do not leave alive.” There have been many programs developed to reduce this rate such as “reducing intake, increasing returns to owner, transport, and adoption” but the rate of the dogs that are dying remained still high (Mohan-Gibbons et al., 2014).

Among the problems that arise with animal shelters, overcrowding is a major factor that deteriorates the well-being of dogs. Overcrowding is said to impose “stress when unfamiliar animals of the same species are mixed, increasing susceptibility to infection, as well as concentrating infectious particles and fomites, and increasing opportunities for transmission to naive or debilitated animals” (Moody, Sen, & Hasan, 2012). Overall, overcrowding makes the animals vulnerable, and can create trauma during the period of being kenneled. These impacts are too significant to be ignored, and re-highlights the importance of reducing the LOS.

An interesting point is that data from American Humane Association (AHA) “reported that more than half (56%) of people” have answered that they would primarily consider getting a dog from a shelter while the people who actually did so were only 22% (Mohan-Gibbons et al., 2014). This difference suggests that there is a bigger pool of people that are willing to acquire

their dogs from the shelter and that an improved shelter management system can aid in heightening the adoption rate of sheltered dogs. The reason might be that people are delaying their decisions to find a dog of their preference, as a study by Garrison and Weiss (2014) showed that “when people were asked what they would do if their source did not have a dog of their choice, almost half (46%) said they would wait or delay their decision” (Garrison & Weiss, 2014). This means that people are willing to wait to find the dog of their preference, so a procedure of identifying attractive traits of the dogs can help in efficient management of the resources that takes to take care of the dogs.

There have been several studies of identifying the characteristics of dogs when people consider adoption. A figure from a research paper written by Garrison and Weiss (2014) suggests that when people consider the acquirement of their dogs, around 36% of people put importance on the source, the place of acquirement, and age and size the following popular factors (Garrison & Weiss, 2014). Another research paper regarding the factors behind adoption decisions shows a different perspective and suggests the dual importance of behavioral and morphological factors. It shows that at first people temporarily choose a dog that they like by observing the morphological factors but examine “behavioral cues to reach” a final decision (Minnis, Davis, & Loftis, 2022). The research inspects from other academic papers that the preferred morphological traits are “younger dogs, smaller-sized dogs, ... healthy dogs” and dogs with “longer coat lengths ... and lighter coat colors” (Minnis et al., 2022).

By referencing the pieces of literature that are mentioned above, this research paper aims to identify the desirable traits of dogs and build a model that predicts the likelihood of adoption by using econometric approaches. It uses data from Kaggle on different dogs' profiles to do so. While the measurement of the dogs' characteristics is limited as some factors such as coat color and dog size don't have a universal standard, the model can still provide a prototype for measuring the likelihood of the adoption, which can be further refined from deepened research.

2. Data and Measurements

This dataset was collected from a well-known data hub Kaggle. It is said that the dataset is a refined version of the original data owned by Mr. Rabie El Kharoua, created to facilitate

easy analysis using various data analysis tools. The time and location of the data measurement were not specified, and the author of the dataset also issues a caution that the data “cannot be used to generalize pet adoption behavior” (El Kharoua, 2024). Despite this, the author states that the data is “collected during a specific period of time,” implying that the dataset is not fabricated and can still provide insights into a new strategy for stray dog shelters (El Kharoua, 2024).

The architecture of the dataset before the data cleansing process consists of 13 columns and 2007 responses. Table 1 shows a brief view of each feature’s description, data type, and example values.

Table 1: Feature Descriptions of the Pet Adoption Dataset

Features	Description	Type	Example Values
PetID	Distinct number that each pet is assigned of	Categorical	534, 2002
PetType	Type of the pet	Categorical	Dog, Cat, Bird, Rabbit
Breed	Classified breed of the pet	Categorical	Golden Retriever, Parakeet
AgeMonths	In months, how old the pet is	Numeric	131, 73
Color	Classified color of the pet	Categorical	White, Orange
Size	Classified size of the pet	Categorical	Small, Medium, Large
WeightKg	In kilograms, the weight of the pet	Numeric	5.28, 29.07
Vaccinated	Whether the pet is vaccinated	Binary	0 – Not vaccinated, 1 – Vaccinated
HealthCondition	Health status of the pet	Binary	0 – Healthy, 1 – Has medical condition
TimeInShelterDays	In days, length of stay at	Numeric	27, 8

	the shelter		
Adoption Fee	In dollars, how many adoption fees are charged	Numeric	43,385
PreviousOwner	Existence of the pet's previous owner	Binary	0 – No, 1 – Yes
AdoptionLikelihood	Pet's final adoption status	Binary	0 – Not adopted, 1 – Adopted

In the cleansed version of the data, only the data that has the PetType of “Dog” remained. Columns that are unnecessary such as PetType and PetID are removed, and a dataset of 522 observations and 11 variables is newly created. Throughout the literature review, it is identified that age, size, and morphological factors play a significant role in people’s decision to adopt a dog, and those are included in the dataset as well. Some additional factors that are not discussed in the literature, such as PreviousOwner, Vaccinated, are also included to see if they show any importance in the decision process.

The point of consideration is that the dataset doesn’t appear to be completely objective, as some data show that two dogs of the same type and size group have completely different weights in kilograms. For example, two dogs that have the ID of 502 and 538 are of the same Golden Retriever type and are classified as Medium. However, the prior dog of 136 months old weighs about 2.08 kg while the later dog of 31 months old weighs about 24.76 kg. This behavior of the collected data is quite consistent, as more odd instances appear in the dataset. An 88-month-old Labrador (PetID: 874) that weighs around 4.65 kg is classified as Large, while an 80-month-old Labrador (PetID: 955) that weighs around 23.47 kg is classified as Small. Considering that they are in the same age range, the standard for classifying the dog that weighs way less than the other one to a bigger size group seems contradictory. An attempt was made to find a different dataset, but as it is hard to gather this type of data, there was no alternative, and a decision was made to stick with the dataset we had.

3. Empirical Model

Because the explained variable has a binary form, logistic regression model was used to yield values between 0 and 1. Yielded value was converted to 0 if it was less than 0.5, and to 1 if it was more than or equal to 0.5. In this way, we were able to classify the values into two categories – Adopted and Not adopted – and calculate the accuracy from it. Different combinations of variables and interaction terms were tried on a generated training set, and the best model was chosen as the one that had high accuracy on a generated test set. The empirical regression model is as follows:

$$\begin{aligned} \text{Logit}(\text{ADOPTIONLIKELIHOOD}_i) = & \beta_0 + \beta_1 \text{BREED_Labrador}_i + \beta_2 \text{BREED_Poodle}_i + \\ & \beta_3 \text{SIZE_Medium}_i + \beta_4 \text{SIZE_Small}_i + \beta_5 \text{BREED_Labrador}_i \times \text{SIZE_Medium}_i + \beta_6 \\ & \text{BREED_Labrador}_i \times \text{SIZE_Small}_i + \beta_7 \text{BREED_Poodle}_i \times \text{SIZE_Medium}_i + \beta_8 \text{BREED_} \\ & \text{Poodle}_i \times \text{SIZE_Small}_i + \beta_9 \text{HEALTHCONDITION}_i + \beta_{10} \text{VACCINATED}_i + \\ & \beta_{11} \text{HEALTHCONDITION}_i \times \text{VACCINATED}_i + \beta_{12} \text{ADOPTIONFEE}_i + \\ & \beta_{13} \text{AGEMONTHS}_i + \beta_{14} \text{ADOPTIONFEE}_i \times \text{AGEMONTHS}_i + \varepsilon_i \quad (1) \end{aligned}$$

4. Results

The result of the OLS Estimation on the train set is shown in Table 2. The high Wald chi-squared value of 70.66 shows that the overall model is statistically significant. The Pseudo R-squared value of 0.4718 is also decent, showing that compared to the null model, about 47% more of the variation in the outcome is explained by the model.

P-values that are less than or equal to 0.05 are marked with an asterisk (*) sign. Low p-value indicates which coefficients are statistically significant, so examining the p-value of the estimators provides a helpful insight into important features of the adoption determination process. Supporting the literature mentioned in the introduction, the estimation result indicates that variables that determine morphological features and age have some importance. Low p-values of the variables BREED_Labrador_i , SIZE_Medium_i , and AGEMONTHS_i that are less than 0.05 provide evidence for it. Outside of the variables that were mentioned in the literature review, the variables VACCINATED_i , $\text{HEALTHCONDITION}_i \times \text{VACCINATED}_i$, ADOPTIONFEE_i were also shown to be significant as the p-values of them are nearly zero.

Overall, the result backs up the findings in the literature while newly identifying that the vaccination status of the dog, the health condition of the dog, and the adoption fee play an important role in the adoption decision process.

Among the three breeds that were in the dataset, the OLS result shows that Labrador is the most preferred dog breed with the highest coefficient, followed by Poodle and Golden Retriever. In terms of size, Medium was the general preference, followed by Small and Large. However, the coefficients of the interaction terms show that the size preference differs by the breed of the dog. For example, large size seemed to be preferred in both Labrador and Poodle breeds since the negative coefficient of the interaction terms $BREED_Labrador_i \times SIZE_Medium_i$, $BREED_Labrador_i \times SIZE_Small_i$, $BREED_Labrador_i \times SIZE_Medium_i$, $BREED_Labrador_i \times SIZE_Small_i$ show that being classified as Medium and Small for those two breeds would lower the chance of adoption. But lower focus should be put on to the interaction terms as their p-value suggests less significance.

The coefficient of $HEALTHCONDITION_i$ is a small negative value, while $VACCINATED_i$ is a positive value. This follows common sense as dogs with health conditions are less appealing than healthy dogs, and dogs that are vaccinated are preferred over unvaccinated dogs. One odd thing is that the interaction term $HEALTHCONDITION_i \times VACCINATED_i$ has a negative coefficient value, and this suggests that when dogs with health conditions are vaccinated, it would lower the likelihood of being adopted. More research seems to be needed to figure out the reasoning behind this value.

The coefficient for $ADOPTIONFEE_i$ and $AGEMONTHS_i$ is a negative value, following the common sense that dogs with higher adoption fees and age are less appealing. The coefficient of the interaction term between these two variables was a positive value, and this suggests that the preference of the older pets may be less swayed by the adoption fees. However, the high p-value of the interaction term suggests less significance, so the focus should be on the individual terms.

Table 2: Estimation Result of the Logistic Regression Model (1)

Number of Observation = 382

Wald $\chi^2(14) = 70.66$

Probability > $\chi^2 = 0.0000$

Pseudo R2 score = 0.4718

* p-value ≤ 0.05

Variable	Coefficient	Robust Standard Error	P-value
BREED_Labrador _i	$\beta_1 = 4.001812$	.764047	0.000 *
BREED_Poodle _i	$\beta_2 = .7640882$	.8261704	0.355
SIZE_Medium _i	$\beta_3 = 4.16239$	.7435575	0.000 *
SIZE_Small _i	$\beta_4 = 1.252411$	.8489807	0.140
BREED_Labrador _i × SIZE_Medium _i	$\beta_5 = -1.794885$	1.019002	0.078
BREED_Labrador _i × SIZE_Small _i	$\beta_6 = -1.385565$	.9527787	0.146
BREED_Poodle _i × SIZE_Medium _i	$\beta_7 = -.9730886$	.9379243	0.300
BREED_Poodle _i × SIZE_Small _i	$\beta_8 = -.1112777$	1.122816	0.921
HEALTHCONDITION _i	$\beta_9 = -.0463575$	1.038588	0.964
VACCINATED _i	$\beta_{10} = 2.940303$	.5086615	0.000 *
HEALTHCONDITION _i × VACCINATED _i	$\beta_{11} = -2.786733$	1.148432	0.015*
ADOPTIONFEE _i	$\beta_{12} = -.004662$	.0021072	0.027*
AGEMONTHS _i	$\beta_{13} = -.0147491$	.0052996	0.005 *
ADOPTIONFEE _i × AGEMONTHS _i	$\beta_{14} = .0000234$	.0000197	0.235
Constant	$\beta_0 = -3.071467$	.8903907	0.001 *

The accuracy analysis of the train and the test set is shown in Table 3. The model can have an overfitting problem when the accuracy of the test dataset is lower than that of the training dataset. However, the empirical model yields a 0.9214 accuracy for the test set and a

lower accuracy of 0.9031 for the train set, showing the unlikeliness of the model having an overfitting problem.

Table 3: Accuracy Analysis of Train Dataset and Test Dataset

Classification	Number of Observations	Accuracy	Standard Deviation
Test Set	140	.9214286	.2700348
Train Set	382	.9031414	.2961531

5. Conclusion

Overall, the OLS result shows support for the literature that considers morphological features as some of the major determinants of adoption decisions. In the model, the features that determine the appearance of the dog, such as breed and size, are shown to be statistically significant with low coefficient p-values. Likewise, the age coefficient has a low p-value and supports the literature's assertion that younger dogs are more likely to get adopted. In addition to these already-known features, adoption fees and vaccination status were newly discovered to influence the adoption decision. Their coefficients and valid p-values show that charging a lower adoption fee and being vaccinated would raise the likelihood of being adopted.

The model scored well on different performance measures such as Wald chi-squared analysis and accuracy analysis, suggesting that it can be a good tool when predicting the likelihood of adoption of sheltered dogs. Through this prediction, the shelters can decide how to use their resources efficiently, how to brand their dogs to the customers, or how to make the dogs more appealing to the customers.

There are some limitations in the study that stem from the data collection process. Since the data collected does not have clear information on when and where it was collected, the situation to which it could be applied is unclear. The data's credibility is even questionable, as the standard for classifying the size of dogs is inconsistent. Moreover, the small number of breed groups further limits the usage of the model, as there are many more than three dog breeds in

the world. Due to this reason, future research on a similar topic should increase the variation of data and have clear standards and specifications that would make the data more credible.

Reference

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No need to answer the following questions for the midterm report.

Q1. What is the point you believe you will receive for this term paper, on a scale of 0 to 100?

Answer: 95

Q2. Give specific reasons for the point you think you will receive.

Answer: I believe that I did a good OLS estimation and interpreted what all the coefficient means to make the readers understand the effect of each variables. I made sure that the model does not contain too much variables to prevent overfitting, and to produce a more generalized model. I wish I had better data that is credible to myself.

Q3. What is the point you believe you will receive for this course, on a scale of 0 to 100?

Why?

Answer: 95